MODULE 01

1.Write a program that generates the Collatz sequence for every number from 1 to N, and prints the number that produces the longest sequence and the maximum value reached across all sequences

```
# Collatz Sequence Analyzer
# For every number from 1 to N:
# - Find which starting number produces the longest Collatz sequence
# - Find the maximum value reached across ALL sequences

def collatz_longest_and_global_max(N):
    memo = {1: 1}        # Cache for sequence lengths
    best_start = 1
    best_length = 1
    global_max_value = 1

    for start in range(1, N + 1):
        n = start
        local_max = n
        path = []

        # Generate Collatz sequence until we hit a memoized number
        while n not in memo:
            path.append(n)
            local_max = max(local_max, n)

            if n % 2 == 0:
                n = n // 2
            else:
                n = 3 * n + 1

            local_max = max(local_max, n)

        # Add lengths back through the stored path
        length = memo[n]
        for value in reversed(path):
            length += 1
            memo[value] = length

        # Update longest sequence info
        seq_length = memo[start]
        if seq_length > best_length:
            best_length = seq_length
            best_start = start
```

```python
        # Update global max value
        global_max_value = max(global_max_value, local_max)

    return best_start, best_length, global_max_value


# Example usage
if __name__ == "__main__":
    N = int(input("Enter N: "))
    start, length, max_value = collatz_longest_and_global_max(N)

    print("Starting number with longest sequence:", start)
    print("Length of that sequence:", length)
    print("Maximum value reached across all sequences:", max_value)
```

OUTPUT:

Enter N: 100

Starting number with longest sequence: 97

Length of that sequence: 119

Maximum value reached across all sequences: 9232

2.Given a paragraph, create a two-dimensional table (26×26) where cell (i, j) stores the number of times character i is followed by character j in the text. Ignore spaces and convert everything to lowercase

```python
# build 26x26 pair-count table for letters (ignore non-letters, lowercase)
def pair_table(text):
    s = ''.join(ch for ch in text.lower() if 'a' <= ch <= 'z')
    table = [[0]*26 for _ in range(26)]
    for a, b in zip(s, s[1:]):
        table[ord(a)-97][ord(b)-97] += 1
    return table

# usage
if __name__ == "__main__":
    txt = input("Enter paragraph: ")
    tbl = pair_table(txt)
    for row in tbl:
        print(' '.join(map(str, row)))
```

OUTPUT:

0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 1 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

3.Read marks of *n* students for *m* sub
Print:

Subject averages